

Antoine EDY

AI Research Scientist

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Paris, France

Education

- **University of Surrey** — *Master of Science in Artificial Intelligence* *2023-2024*
  - International Excellence Scholarship recipient. Graduated with Highest Honors
  - NLP, CV, Audio Processing, Symbolic Knowledge Representation and Logical Reasoning, People-Centered AI
  - Master Thesis: *Zero-shot and Few-shot Adaptation of Multimodal Foundation Models*
- **École Centrale de Lyon** — *General Engineering Program (Grande École)* *2021-2024*
  - Specializations: Mathematics, Machine Learning, Probability Theory, Chaos and Fractals
  - Project Leader: Unsupervised MCU Classification with Deeplearning for STMicroelectronics
- **Collège Stanislas Paris VI** — *Classe Préparatoire, MPSI-MP* *2019-2021*

Experience

- **Illuin Technology** — *AI Research Scientist* *September 2024 - Present*
  - Co-led the creation of ViDoRe V3, establishing the gold standard benchmark for multimodal document retrieval evaluation; contributor to the MTEB ecosystem
  - Designed an agentic harness that writes, stores and retrieves memory for continuous in-context learning on long-horizon tasks
  - Conducted research on Multimodal Retrieval and late interaction dynamics (ColBERT/ColPali-like models)
  - Fine-tuned LLMs for tool calling in agentic RAG contexts; fine-tuned speech-to-text models (Whisper, NeMo) for domain-specific industrial applications
- **Centre for Vision, Speech and Signal Processing (University of Surrey)** — *Research Intern* *2024*
  - Studied zero-shot semantic segmentation using multimodal foundation models; improved ZegCLIP baseline with multi-scaling techniques and inner encoder layers
  - Research led to Master's thesis: *Zero-shot and Few-shot Adaptation of Multimodal Foundation Models*, supervised by Dr Muhammad Awais Rana
- **AI-vidence** — *ML Research Intern* *2023*
  - Developed a python library for tabular model explainability and cluster analysis
  - Presented at SpringTech Paris-Saclay 2023 at HEC Paris

Publications

- **ViDoRe V3: A Comprehensive Evaluation of RAG in Complex Real-World Scenarios**
  - A. Loison⋆, Q. Macé⋆, **A. Edy**⋆, V. Xing, T. Balough, G. Moreira, B. Liu, M. Faysse et al.
  - Multimodal RAG benchmark: 26,000 pages, 18,594 queries across 10 domains in 6 languages, 12,000h of human annotation. Work made in collaboration with NVIDIA.
  - **ACL 2026 Main Conference** (Oral Presentation), arXiv:2601.08620
- **Working Notes on Late Interaction Dynamics: Analyzing Targeted Behaviors**
  - **A. Edy**⋆, M. Conti⋆, Q. Macé
  - Analysis of length bias in multi-vector scoring and MaxSim similarity distribution on NanoBEIR benchmark
  - **ECIR 2026** Late Interaction Workshop, arXiv:2603.26259
- **Exploring Zero-Shot Capabilities of Multi-Modal Foundation Models for Semantic Segmentation**
  - Improved ZegCLIP baseline using multi-scaling techniques and inner encoder layers for zero-shot segmentation
  - **Master's Thesis**, University of Surrey, ResearchGate

Talks & Blog Posts

- ViDoRe V3 and Visual Retrieval at Neurons & Peppers, Invited talk at Datacraft Paris *2026*
- Hugging Face blog post author on agentic search and evaluation, Hugging Face
- Late Interaction Dynamics at ECIR 2026, Late Interaction Workshop *2026*

Open Source Contributions

- Contributor to MTEB, colpali-engine, vidore-benchmark, widgetti apps
- Developer of IStacky, an open-source Python image-processing library on PyPI GitHub

Technical Skills

Deep Learning: PyTorch, Hugging Face (Transformers, PEFT), NVIDIA NeMo, OpenMMLab
Retrieval & NLP: ColBERT/ColPali (late interaction), MTEB, Sentence-Transformers, FAISS, BM25
Languages & Tools: Python (advanced), C/C++, SQL, Bash, LaTeX, Git
Spoken: French (Native), English (C1, TOEFL 623/677), Spanish (B1)